

Pharmacokinetics

Pimecrolimus and Tacrolimus

Tacrolimus ointment, 0.03% and 0.1%, applied twice daily, was studied extensively in clinical trials involving adults and children as young as 2 years of age. Pharmacokinetic studies showed percutaneous absorption in 10% to 20% of patients with atopic dermatitis initially, but blood levels generally became undetectable within 1 week as skin healing and barrier function were restored. Blood concentrations rarely exceeded 2 ng/mL, compared to trough levels of 5–15 ng/mL in transplant patients receiving systemic tacrolimus.

In an open-label study of twice-daily pimecrolimus use in infants aged 5.7 to 11.9 months at baseline with moderate to severe atopic dermatitis, Lekhanpaul et al. found no drug accumulation and systemic levels ranging from below the limit of detection to 1.94 ng/mL. No significant adverse events were reported during the 1-year trial.

Imiquimod

The half-life of systemically administered imiquimod (oral or subcutaneous) is 2 to 4 hours in healthy volunteers. Following intravenous administration, the half-life extends to 14 hours. After a single topical application, less than 1% of the drug is excreted in the urine. The primary systemic metabolite, S26704, is a hydroxylated derivative that reaches plasma concentrations less than 20% of the parent compound. Higher systemic levels have been observed with intravaginal administration in healthy volunteers or systemic administration in chronically ill patients. With topical application in healthy volunteers, imiquimod levels are minimal (measured in pg/mL), and the metabolite S26704 is not detectable.

Indications: U.S. Food and Drug Administration